Supplementary Materials for 'A Bio-inspired Spatial-Frequency Synergistic Model for Small Object Detection'

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Abstract

In the Supplementary Materials, we provide additional experimental data, visualizations, and experimental results of BFSCDet that were not fully presented in the main manuscript, including stability analysis, generalization experiments, results and analysis of several ablation studies, and failure cases.

Keywords: Biologically inspired, small object detection, aerial images, spatial-frequency synergistic.

1. Stability Analysis

To assess model stability, repeatability experiments of BFSCDet were conducted on the VisDrone dataset. As shown in Fig.1, under the same experimental setup, four repeated trials yielded a sample-unbiased variance of 3.67×10^{-6} for mAP50 and 3.3×10^{-7} for mAP50-95, indicating minimal deviation. The coefficients of variation for the two metrics were 0.39% and 0.19%, both below 1%, demonstrating very low fluctuation in accuracy metrics, high network robustness, and reliable results.

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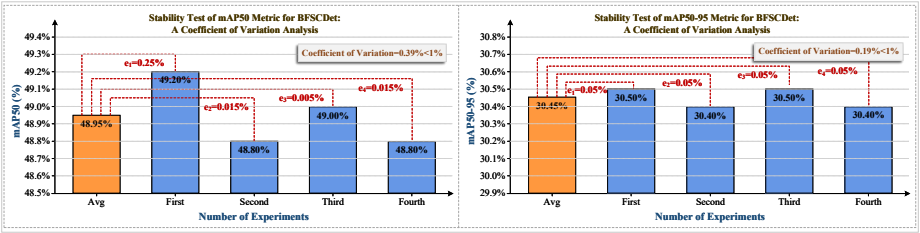

Fig.1. Visualization of BFSCDet Stability Analysis Results.

Table 1. Performance results in VisDrone2019 for different fusion strategies applied at various layers in SFF.

Layer1	Layer2	Layer3	Layer4	mAP50	mAP50-95	GFLOP	Param(M)
SDC_res	SDC_res	SDC_res	SDC_res	47.4	29.2	45.7	1.82
SDC_res	SDC_res	SDC_res	SDC_cat	47.6	29.3	46.0	1.87
SDC_res	SDC_res	SDC_cat	SDC_res	48.0	29.7	46.1	1.85
SDC_cat	SDC_cat	SDC_cat	SDC_cat	48.2	29.8	47.5	1.94
SDC_cat	SDC_res	SDC_cat	SDC_cat	48.2	29.6	47.2	1.93
SDC_cat	SDC_cat	SDC_res	SDC_cat	47.9	29.6	47.2	1.92
SDC_cat	SDC_cat	SDC_cat	SDC_res	47.9	29.5	47.8	1.93
SDC_res	**SDC_cat**	**SDC_cat**	**SDC_cat**	**48.6**	**30.2**	**47.8**	**1.93**

2. Ablation study of the two spatial-frequency fusion methods

As shown in Table 1, we evaluated two fusion strategies-residual connection (SDC_res) and feature concatenation (SDC_cat), applied to the outputs of HSE and LGE at different layers. When SDC_res in the first layer and SDC_cat in the remaining layers achieved the best results. The advantage of this design lies in its residual connections, which fully preserve shallow features and prevent early-stage feature degradation. In subsequent layers, multi-scale convolutions strengthen spatial information processing, forming an effective complementarity with frequency-domain features. This synergy enhances network accuracy, with 48.6% mAP50 and 30.2% mAP50-95.

3. Ablation study of the SFF module

For the SFF module, we design eight ablation configurations by evaluating different combinations of OnConv and OffConv, which simulate the receptive fields of parvocellular and magnocellular pathways, together with the spatial module SDC. During the experiments, OnConv and OffConv are replaced with standard 3×3 convolutions when disabled. As reported in Table 2, the combination of OffConv and OnConv with

the spatial–frequency fusion module SDCE achieves the best performance, reaching 47.7 mAP50 and 29.5 mAP50-95. These results indicate that modeling parvocellular receptive fields to capture high-frequency information enhances fine-grained object details, while magnocellular large receptive fields capturing low-frequency components effectively preserve global structure. When further combined with multi-scale spatial receptive fields, object saliency is strengthened, leading to improved detection accuracy.

Table 2. Ablation study results of each module in SFF on the VisDrone2019 dataset.

OnConv	OffConv	SDC	mAP50	mAP50-95	GFLOP	Param(M)
			45.6	27.9	40.2	1.44
✓			47.2	29.1	47.6	1.64
	✓		47.0	29.0	47.8	1.65
		✓	46.2	28.3	47.6	1.69
✓	✓		47.6	29.4	47.8	1.75
✓		✓	47.6	29.4	47.7	1.75
	✓	✓	47.3	29.2	47.8	1.93
✓	✓	✓	**47.7**	**29.5**	**48.7**	**1.77**

4. Ablation study of the VIFM-SE module

Another set of experiments focuses on the VIFM-SE module, where detection performance under channel enhancement applied to different feature maps is compared, as shown in Table 3. Specifically, three configurations are evaluated: channel enhancement applied only to shallow feature maps, only to deep feature maps, and to both feature maps. The results show that enhancing only one of the two yields an mAP50 of 46.2%, with limited reduction in parameter count. In contrast, applying self-enhancing channel attention to both feature maps improves mAP50 by 0.5 and mAP50-95 by 0.3, demonstrating that the proposed channel attention mechanism effectively strengthens small-object representation while suppressing background interference in complex scenes, thereby improving detection performance.

Table 3. Ablation study results each module in VIFM-SE in VisDrone2019.

VIFM-SE-H	VIFM-SE-L	mAP50	mAP50-95	GFLOP	Param(M)
		45.6	27.9	40.1	1.41
✓		46.2	28.1	38.8	1.46
	✓	46.2	28.4	38.8	1.45
✓	✓	**46.7**	**28.7**	**40.2**	**1.49**

5. Generalization Experiments

To evaluate the generalization ability of BFSCDet in non-aerial scenarios, we performed cross-domain task transfer. Specifically, experiments were conducted on the ground-view TinyPerson dataset, the marine organism dataset iBISH, and the medical dataset Brain Tumor. As shown in Table 4, compared to YOLOv5 and YOLOv8, BFSCDet maintains superior accuracy, highlighting its strong adaptability in challenging marine environments and in detecting tumors with blurred boundaries.

Table 4. Generalization experimental results comparison (bold indicates the best performance)

	TinyPerson		iBFISH		Brain Tumor		
Method	**mAP50**	**mAP50-95**	**mAP50**	**mAP50-95**	**mAP50**	**mAP50-95**	**Params**
YOLOv5	19.1	4.80	37.6	18.2	50.2	36.7	2.29
YOLOv8	18.1	6.59	37.5	18.0	51.0	38.1	2.79
BFSCDet	**28.2**	**9.44**	**46.1**	**25.1**	**51.3**	**36.8**	**1.95**

As shown in Fig.2 , the visualization results demonstrate that BFSCDet successfully detects most objects in ground-view scenes. In aquatic environments, where fish appear blurred and are affected by cluttered marine backgrounds, BFSCDet still identifies and classifies objects effectively. Moreover, it accurately recognizes brain tumors despite their ambiguous boundaries and limited class diversity.

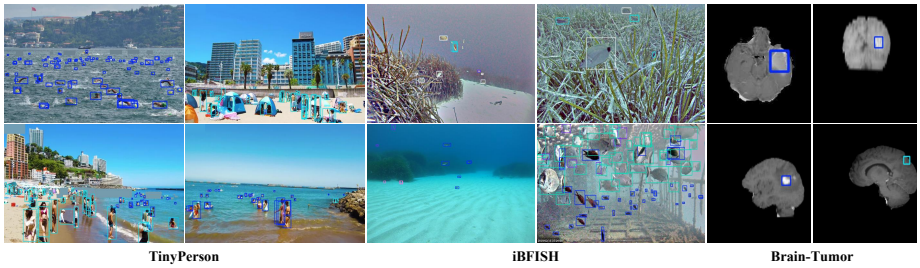

Fig.2. Visualization Results of BFSCDet on Non-Aerial Datasets

6. Visualization Results of BFSCDet-n on Three Small Object Datasets

To demonstrate that the lightweight BFSCDet-n achieves comparable detection performance to the enhanced BFSCDet-l for small objects under diverse scales, motion blur, and limited illumination, we test BFSCDet-n on the same scene images from the three datasets in Fig.8 in Experiment. As shown in Fig.3, its detection results are highly similar to BFSCDet-l, achieving equally excellent performance.

Fig.3. Visualization of BFSCDet-n detection results on the VisDrone2019, UAVDT, and VEDAI.

7. Failure Cases

The proposed biologically inspired spatial-frequency synergistic network BFSCDet achieves superior detection performance in scenarios with diverse object scales, poor illumination, motion blur, and complex backgrounds. However, the model still has failure cases.

First, it is prone to false detections in scenes where objects exhibit highly similar features. As shown in Fig.4(a), since the categories “pedestrians” and “People” both

represent humans with highly similar features, BFSCDet confuses them during detection. Similarly, as indicated by the red box in (b), the network misclassifies a bicycle as a motorbike.

Second, the model suffers from duplicate detections for some objects in highly crowded scenes. As shown in (c), the model misdetects a van as a car and produces multiple bounding boxes on the same object. Furthermore, only four cars appear in the red box of (d), but the model generates multiple duplicate boxes for them.

To address these issues, we will improve the detection head of the network by incorporating the encoding and integration characteristics of the inferotemporal (IT) cortex in the visual system, enabling accurate detection in crowded and high-similarity scenarios.

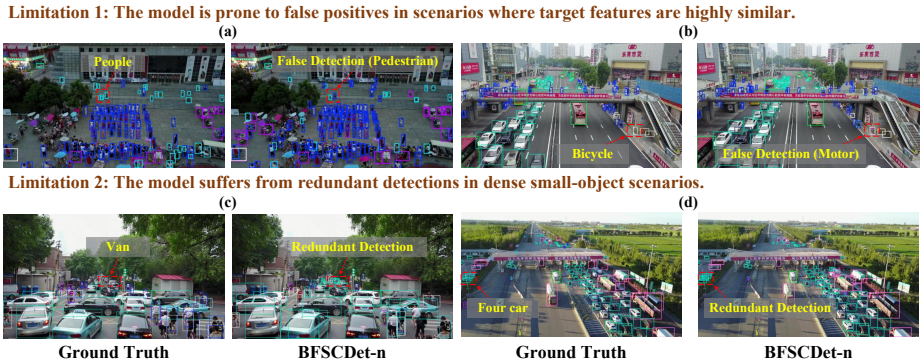

Fig.4. Failure Cases of BFSCDet in Detection